

SYDNEY GIRLS HIGH SCHOOL



YEAR 12 MATHEMATICS
Extension 1
2012

ASSESSMENT TASK 1

November 2011

Time allowed: 60 minutes +5 min reading

Integration and Locus

Instructions:

- There are Five (5) questions. Questions are of equal value.
- Attempt all questions.
- Show all necessary working. Marks may be deducted for badly arranged work.
- Start each question on a new page. Write on one side of the paper only.
- Note: $\int x^n dx = \frac{1}{n+1} x^{n+1}$, $n \neq -1$; $x \neq 0$, if $n < 0$

Name:

Teacher's Name

QUESTION ONE (12 marks)

a) Given $f'(x) = 3x^2 - 3x + 1$, find the equation of the curve which passes through $(-1, 0)$. (2)

b) Find

i) $\int \frac{x^3 + 2x}{x^3} dx$ (2)

ii) $\int x(x-1)^2 dx$ (2)

c) A and B are the points $(1, 3)$ and $(5, 9)$ respectively. The point $P(x, y)$ moves so that $m_{PA} \times m_{PB} = -1$. Find the locus of point P . (3)

d) The parabola $x^2 = ky$ passes through the point $(-6, 3)$. Find

i) The coordinates of the focus (2)

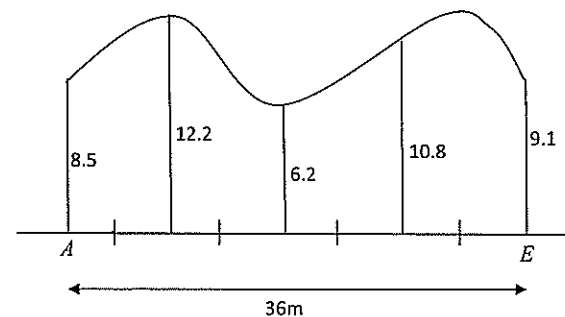
ii) The equation of the directrix (1)

QUESTION TWO (12 marks)

a) Find the area bounded by $y = 3x - x^2$ and $y = 3 - x$. (4)

b) Find $\int \frac{dx}{(5x+3)^2}$ (2)

c) Use Simpson's rule to find an approximation of the area between the road and fence between A and E . (3)



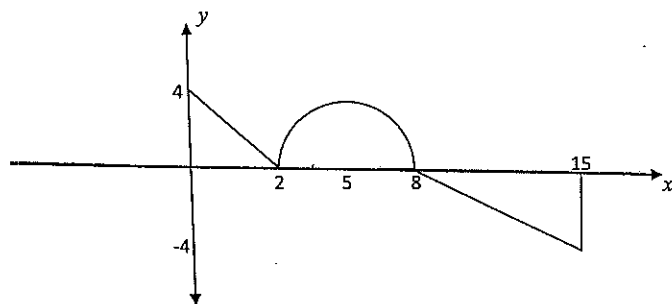
d) Evaluate

$$\int_1^4 \left(x - \frac{2}{x}\right)^2 dx \quad (3)$$

QUESTION THREE (12 marks)

a) Find $\int \frac{x^3}{\sqrt{x}} dx$ (2)

b) The graph below shows $y = f(x)$. Evaluate $\int_0^{15} f(x) dx$ (2)



c) The point $P(x, y)$ moves so that $\frac{PO}{PA} = \frac{2}{1}$, where O is the origin and $A(3, 0)$.
 i) Find the locus of P (3)
 ii) Describe this locus (1)

d) A vase is designed to hold $50\pi \text{ cm}^3$ of water. Its shape is determined by rotating the parabola $x = \frac{y^2}{30}$ in the first quadrant about the y axis. If the height of the vase is $k \text{ cm}$,

i) Show that $\int_0^k y^4 dy = 45000$ (2)
 ii) Find the value of k (correct to the nearest whole number) (2)

QUESTION FOUR (12 marks)

a) For the parabola $y = x^2 - 6x + 4$ find

- i) The coordinates of the vertex (2)
- ii) The coordinates of the focus (1)
- iii) The equation of the directrix (1)

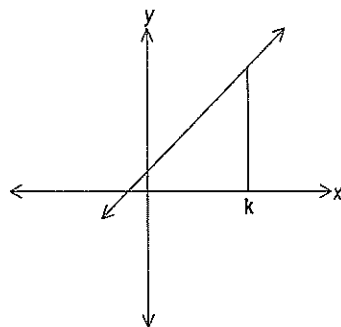
b) Find the area between the curve $y = x^2 - x - 6$ the x axis, and the lines $x = 0$ and $x = 4$. (3)

c) Evaluate $\int_0^4 -\sqrt{16-x^2} dx$ (2)

d) Sketch the parabola $(x+1)^2 = 6(6-2y)$ showing the co-ordinates of the focus and the equation of the directrix. (3)

QUESTION FIVE (12 marks)

- a)
- i) Find the equation of the tangent to the parabola $x^2 = 4y$ at $x = 4$. (2)
 - ii) Find the area enclosed by the curve, the tangent and the x axis. (3)
- b) Find the volume of the solid of revolution if the area bounded by the curves $y = x^2$ and $y = 4 - x^2$ rotates around the y axis. (3)
- c) The shaded area in the diagram is bounded by the x axis, the y axis, the line $y = 5x + 1$ and the $x = k$. This area equals to $44 u^2$.
- i) Find the value of k (2)
 - ii) This area is rotated about the y axis. Find the volume of the solid generated. (2)



THE END

Year 12, Ext 1 2011

1) a)

$$f(x) = \frac{3x^3}{8} - \frac{3x^2}{2} + x + C$$

$$0 = -1 - \frac{3}{2} - 1 + C$$

$$0 = -\frac{7}{2} + C$$

$$C = \frac{7}{2}$$

$$f(x) = x^3 - \frac{3x^2}{2} + x + \frac{7}{2}$$

b) i) $\int 1 + \frac{2}{x^2} dx$

$$\int 1 + 2x^{-2} dx$$

$$= x + \frac{2x^{-1}}{-1} + C$$

$$= x - \frac{2}{x} + C$$

ii) $\int x(x^2 - 2x + 1) dx$

$$\int x^3 - 2x^2 + x$$

$$= \frac{x^4}{4} - \frac{2x^3}{3} + \frac{x^2}{2} + C$$

c) $\frac{y-3}{x-1} \times \frac{y-9}{x-5} = 1$

$$\frac{y^2 - 9y - 3y + 27}{x^2 - 5x - x + 5} = 1$$

$$y^2 - 12y + 27 = -x^2 + 6x - 5$$

$$x^2 - 6x + y^2 - 12y + 32 = 0$$

$$x^2 - 6x + 9 + y^2 - 12y + 36 = -32 + 45$$

$$(x-3)^2 + (y-6)^2 = 13$$

d) $x^2 = ky$

$$36 = k \times 3$$

$$k = 12$$

$$4a = 12$$

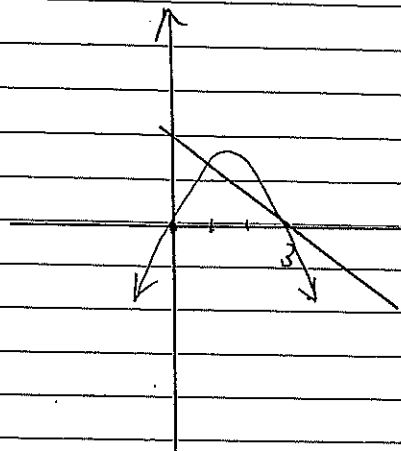
$$a = 3$$

i) $F(0, 3)$

ii) $y = -3$

Q2

a)



$$y = x(3-x)$$

$$y = 3-x$$

$$3-x = 3x-x^2$$

$$x^2 - 4x + 3 = 0$$

$$(x-3)(x-1) = 0$$

$$x = 3 \text{ or } 1$$

$$A = \int_1^3 3x - x^2 (3-x) dx$$

$$= \int_1^3 3x - x^2 - 3 + x dx$$

$$= \int_1^3 4x - x^2 - 3 dx$$

$$= \left[\frac{4x^2}{2} - \frac{x^3}{3} - 3x \right]_1^3$$

$$= [18 - 9 - 9] - [2 - \frac{1}{3}]$$

$$= -\left(-\frac{4}{3}\right)$$

$$= \frac{4}{3} \text{ m}^2$$

b) $\int (5x+3)^{-2}$

$$= \frac{(5x+3)^{-1}}{-1 \times 5} + C$$

$$= -\frac{1}{5(5x+3)} + C$$

c) $A = \frac{9}{3} \times 122$

$$= 366 \text{ m}$$

x	f(x)	w	f(x)w
0	8.5	1	8.5
9	12.2	4	48.8
18	6.2	2	12.4
27	10.8	4	43.2
36	9.1	1	9.1

$$\Sigma = 122$$

d) $\int_1^4 x^2 - 4 + 4x^{-2} dx$

$$= \left[\frac{x^3}{3} - 4x - \frac{4x^{-1}}{1} \right]_1^4$$

$$= \left[\frac{x^3}{3} - 4x - \frac{4}{x} \right]_1^4$$

Q3

$$\begin{aligned}
 a) \int x^3 \times x^{-\frac{1}{2}} dx \\
 &= \int x^{\frac{5}{2}} dx \\
 &= \left[\frac{2x^{\frac{7}{2}}}{\frac{7}{2}} + C \right]
 \end{aligned}$$

$$\begin{aligned}
 b) \int_0^{15} f(x) dx &= \frac{1}{2} 4 \times 2 + \frac{1}{2} \pi \times 3^2 - \frac{1}{2} 4 \times 7 \\
 &= 4 + \frac{9\pi}{2} - 14 \\
 &= \frac{9\pi}{2} - 10
 \end{aligned}$$

c) i) PO = 2PA

$$\sqrt{(x-0)^2 + (y-0)^2} = 2\sqrt{(x-3)^2 + (y-0)^2}$$

$$\sqrt{x^2 + y^2} = 2\sqrt{(x-3)^2 + y^2}$$

$$x^2 + y^2 = 4[(x^2 - 6x + 9) + y^2]$$

$$x^2 + y^2 = 4x^2 - 24x + 36 + 4y^2$$

$$3x^2 + 3y^2 - 24x + 36 = 0$$

$$x^2 + y^2 - 8x = -12$$

$$x^2 - 8x + 16 + y^2 = 12 + 16$$

$$(x-4)^2 + y^2 = 4$$

ii)

circle

C(4,0)

r = 2

d)

$$V = \pi \int \left(\frac{y^2}{30} \right)^2 dy$$

$$50\pi = \pi \int_0^k \frac{y^4}{900} dy$$

$$\int_0^k y^4 dy = 900 \times 50$$

$$\int_0^k y^4 dy = 45000$$

$$\left[\frac{y^5}{5} \right]_0^k = 45000$$

$$\frac{k^5}{5} = 45000$$

$$k^5 = 225000$$

$$k = 12$$

Q4) a)

$$y = x^2 - 6x + 4$$

$$y - 4 = x^2 - 6x$$

$$x^2 - 6x + 9 = y - 4 + 9$$

$$(x-3)^2 = y + 5$$

$$V(3, -5)$$

$$4a = 1$$

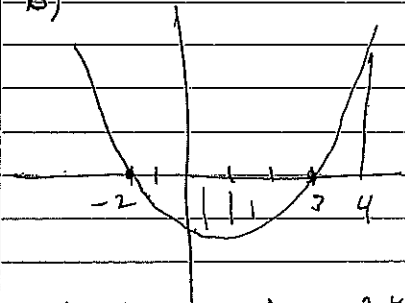
$$a = \frac{1}{4}$$

$$F\left(3, -4\frac{3}{4}\right)$$

$$y = -5\frac{1}{4}$$

$$(x-3)(x+2)$$

b)



$$\begin{aligned}
 A &= \left| \int_0^3 x^2 - x - 6 \right| + \int_3^4 x^2 - x - 6 \\
 &= \left| \left[\frac{x^3}{3} - \frac{x^2}{2} - 6x \right]_0^3 \right| + \left[\frac{x^3}{3} - \frac{x^2}{2} - 6x \right]_3^4 \\
 &= \left| \left(\frac{27}{3} - \frac{9}{2} - 18 \right) \right| + \left(\frac{64}{3} - 8 - 24 \right) - \left(\frac{27}{3} - \frac{9}{2} - 18 \right)
 \end{aligned}$$

$$= \frac{27}{2} + \frac{17}{6}$$

$$= \frac{49}{3}$$

$$= 16\frac{1}{3} \text{ u}^2$$

$$c) \int_0^4 -\sqrt{16-x^2} dx$$

$$= -\frac{1}{4} \pi \times 4^2$$

$$= -4\pi$$

$$\text{at } x=4 \quad y=4$$

$$y-4 = 2(x-4)$$

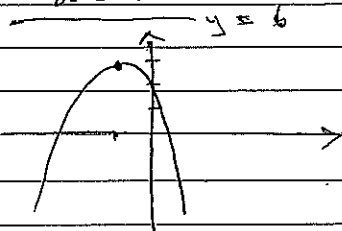
$$y = 2x - 8 + 4$$

$$\boxed{y = 2x - 4}$$

$$d) (x+1)^2 = -12(y-3)$$

$$4a = 12$$

$$a = 3$$



$$V = (-1, 3)$$

$$F = (-1, 0)$$

Directrix

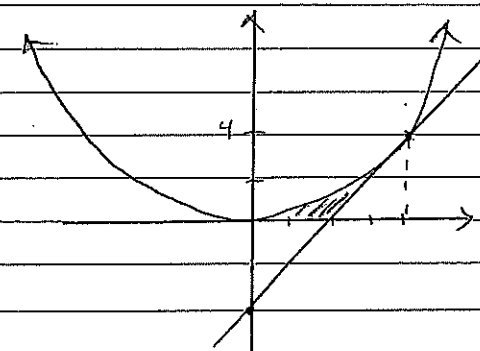
$$y = 6$$

5) a)

$$y = \frac{x^2}{4}$$

$$y' = \frac{2x}{4} = \frac{x}{2}$$

$$\text{at } x=4 \quad m=2$$



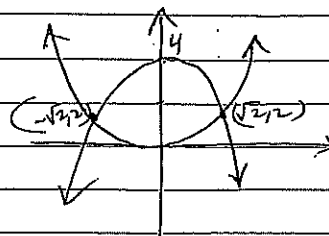
$$A = \int_0^4 \frac{x^2}{4} dx - \frac{1}{2} \times 4 \times 2$$

$$= \left[\frac{x^3}{12} \right]_0^4 = 4$$

$$= 5\frac{1}{3} - 4$$

$$= 1\frac{1}{3}$$

b)



$$x^2 = 4 - y^2$$

$$2x^2 = 4$$

$$x^2 = 2$$

$$x = \pm\sqrt{2}$$

$$x = \sqrt{2} \quad \therefore y = 2$$

$$V = \pi \int_0^2 y dy + \pi \int_2^4 (4-y) dy$$

$$= \pi \left[\frac{y^2}{2} \right]_0^2 + \pi \left[4y - \frac{y^2}{2} \right]_2^4$$

$$= 2\pi + \pi (16 - 8 - (8 - 2))$$

$$= 2\pi + \pi (2)$$

$$= 4\pi \text{ u}^3$$

c)

$$i) A = \int_0^k 5x + 1 dx$$

$$44 = \left[\frac{5x^2}{2} + x \right]_0^k$$

$$44 = \frac{5k^2}{2} + k$$

$$88 = 5k^2 + 2k$$

$$5k^2 + 2k - 88 = 0$$

$$5k^2 + 22k - 20k - 88 = 0$$

$$k(5k + 22) - 4(5k + 22) = 0$$

$$(k-4)(5k+22) = 0$$

$$k = 4 \text{ or } -\frac{22}{5}$$

$$k = 4 \rightarrow y$$

ii)

$$V = \pi \times 4^2 \times 21$$

$$\pi \int_1^{21} \left(\frac{y-1}{5} \right)^2 dy$$

$$= 336\pi - \pi \left[\frac{(y-1)^3}{3} \right]_1^{21}$$

$$= 336\pi - 106\frac{2}{3}\pi$$

$$= 229\frac{1}{3}\pi$$